THE UNIVERSITY OF MELBOURNE

Marketing Research

Week 11

Recap

1. Check the regression analysis data requirements
2. Specify and estimate the regression model
3. Test the regression analysis assumptions
4. Interpret the regression results
5. Validate the regression results
6. Use the regression model

Levels	D1	D2
Blue (1)	0	0
Silver (2)	**1**	0
Gold (3)	0	**1**

Topics for today

1. Data visualisation
2. Principal Component & Factor Analysis
3. Group work day on your group project

Data Visualization

Data Visualization

Importance of Data Visualization

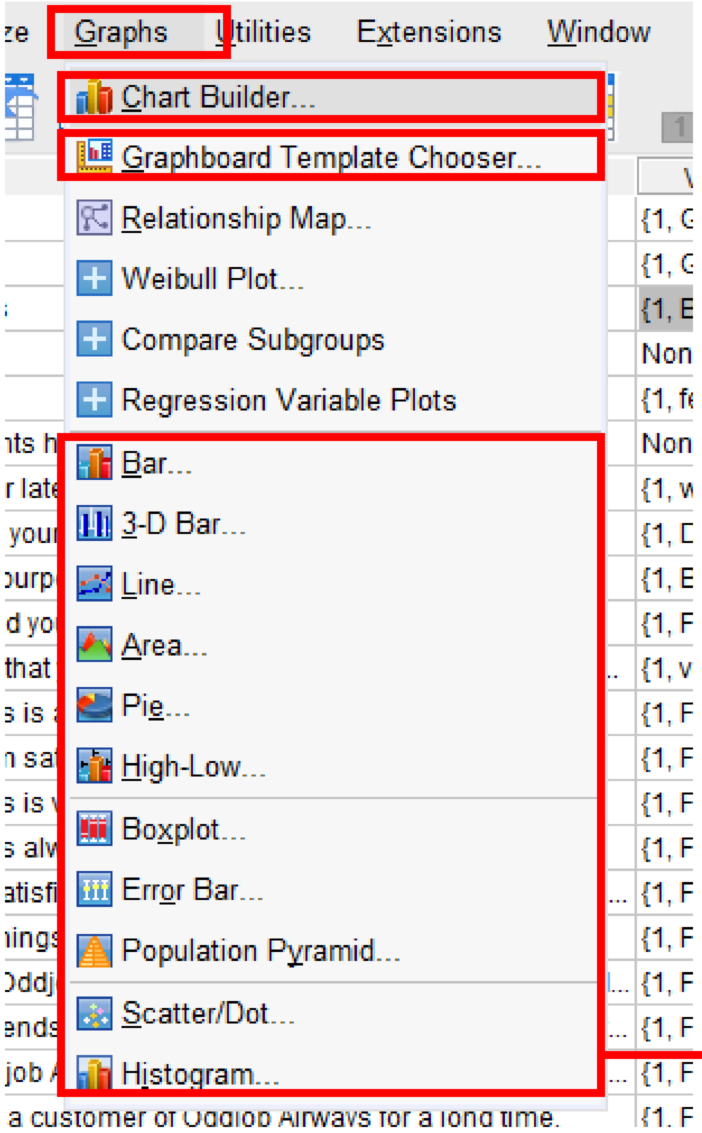
- Enable digestion of information for swift, quick actions
- Spot connections between different variables
- Observe emerging trends
- Tell a story of your analysis with visualization of data
- Engage audience and encourage discussion

Data Visualization Tools

Excel, Tableau Public (Free), SPSS, HighCharts (free license for personal use), Qlikview, Sisense etc..

Data Visualization

Graphic Options in SPSS



or *Legacy Dialogs*

Choosing the most Appropriate Chart

Chart Type	Best For
Bar chart	Categorical comparisons
Line chart	Trends over time
Pie chart	Proportions (use sparingly)
Scatterplot	Relationships between two variables
Histogram	Displaying distribution of a single metric variable

Examples:

•Use *bar charts* for survey responses (e.g., “preferred brand”).

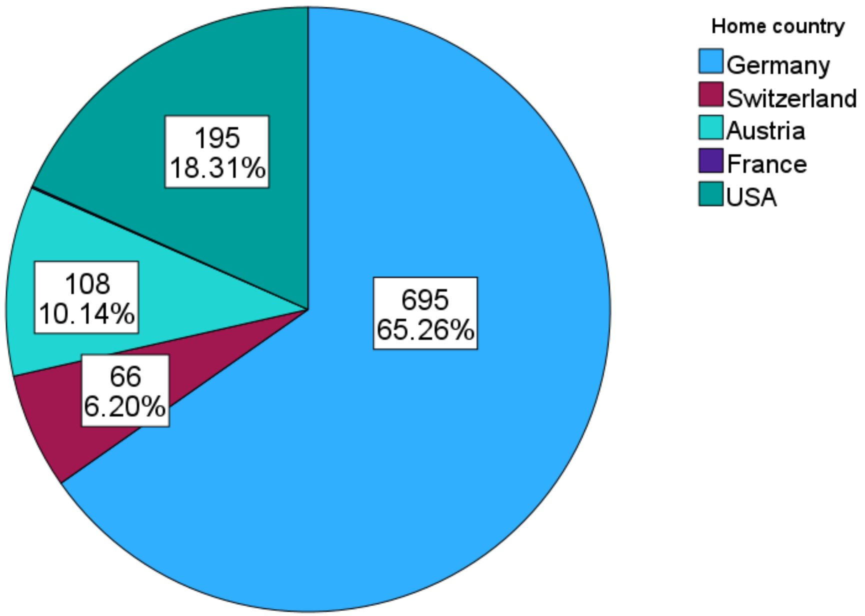
•Use *histograms* to show the distribution of continuous variables (e.g., age, purchase amount).

•Avoid *3D pie charts*—they distort perception and add no value.

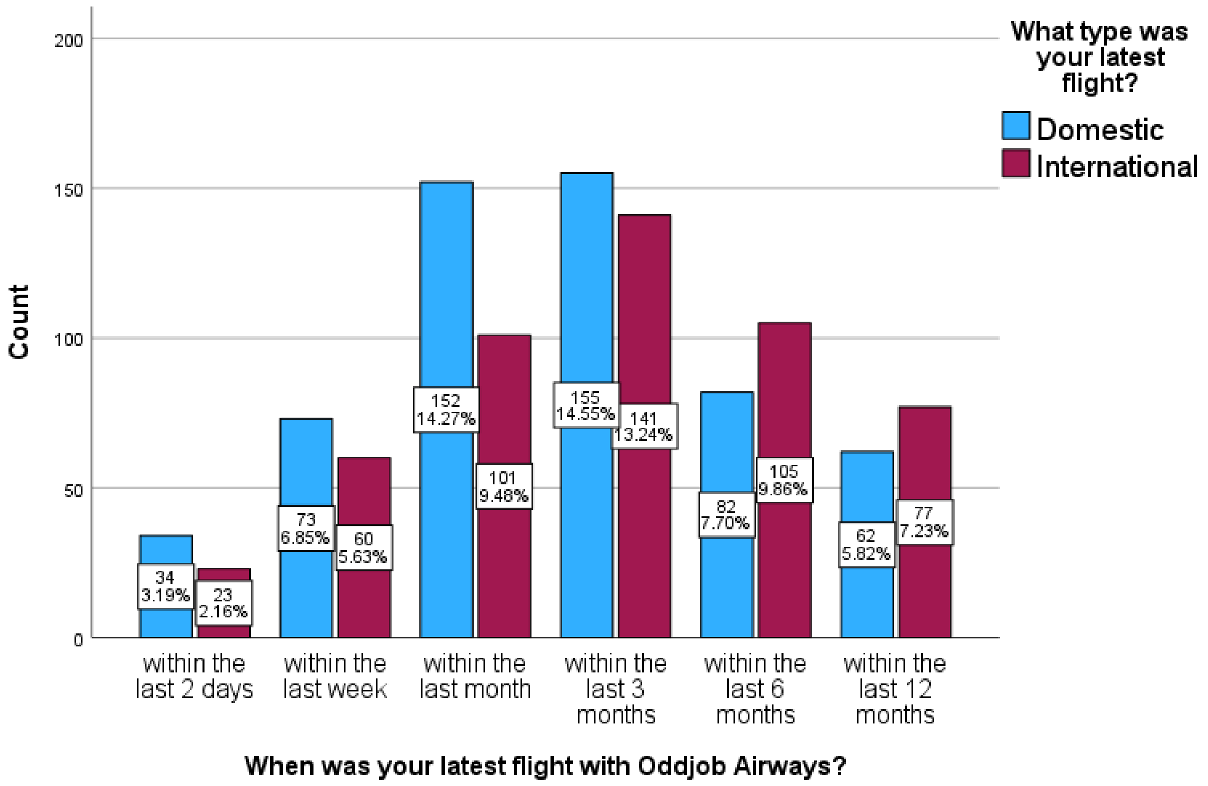
Data Visualization for Categorical Variables

Categorical Variables (nominal and ordinal)

- Bar
- Pie



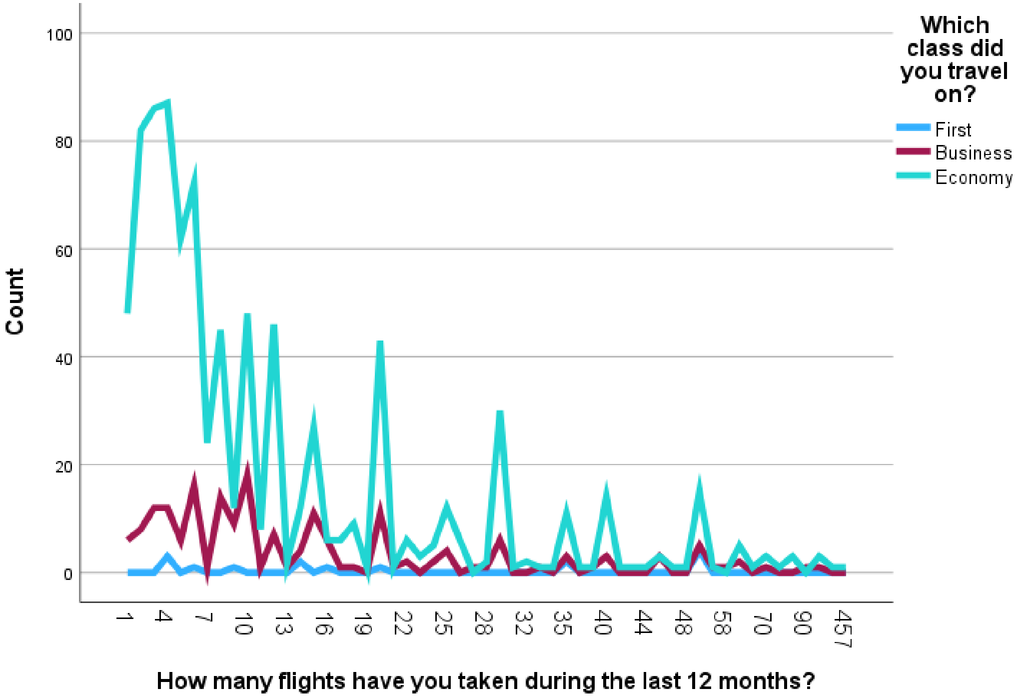
Define Slices by - Country

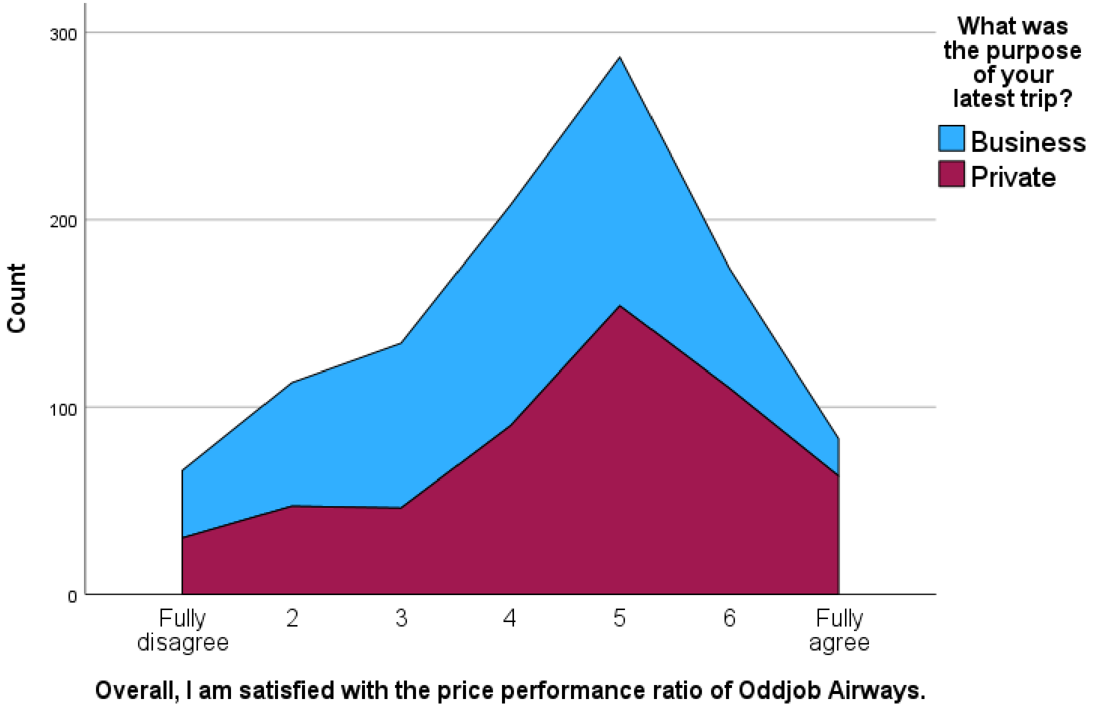


Category axis – flight_latest
Define clusters by – flight_type

Data Visualization for Metric Variables

- Line
- Area



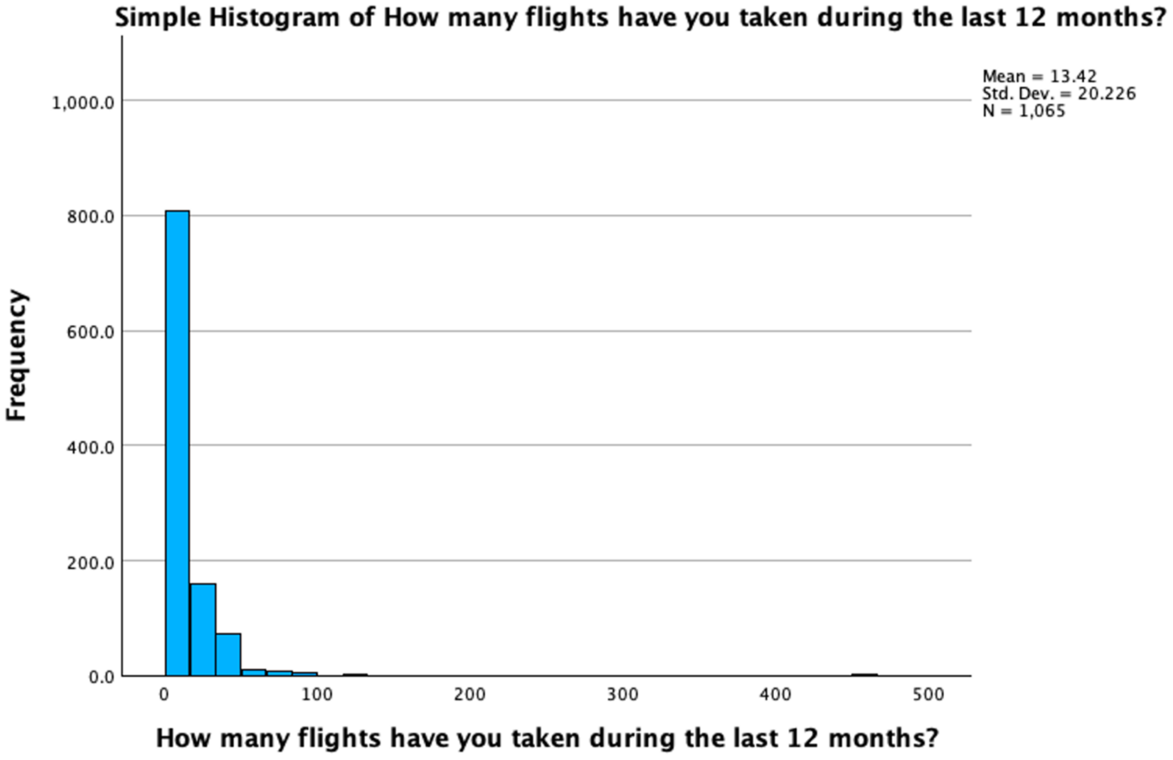


Category axis – nflights
Define lines by – flight_class

Category axis – overall_sat
Define areas by – flight_purpose

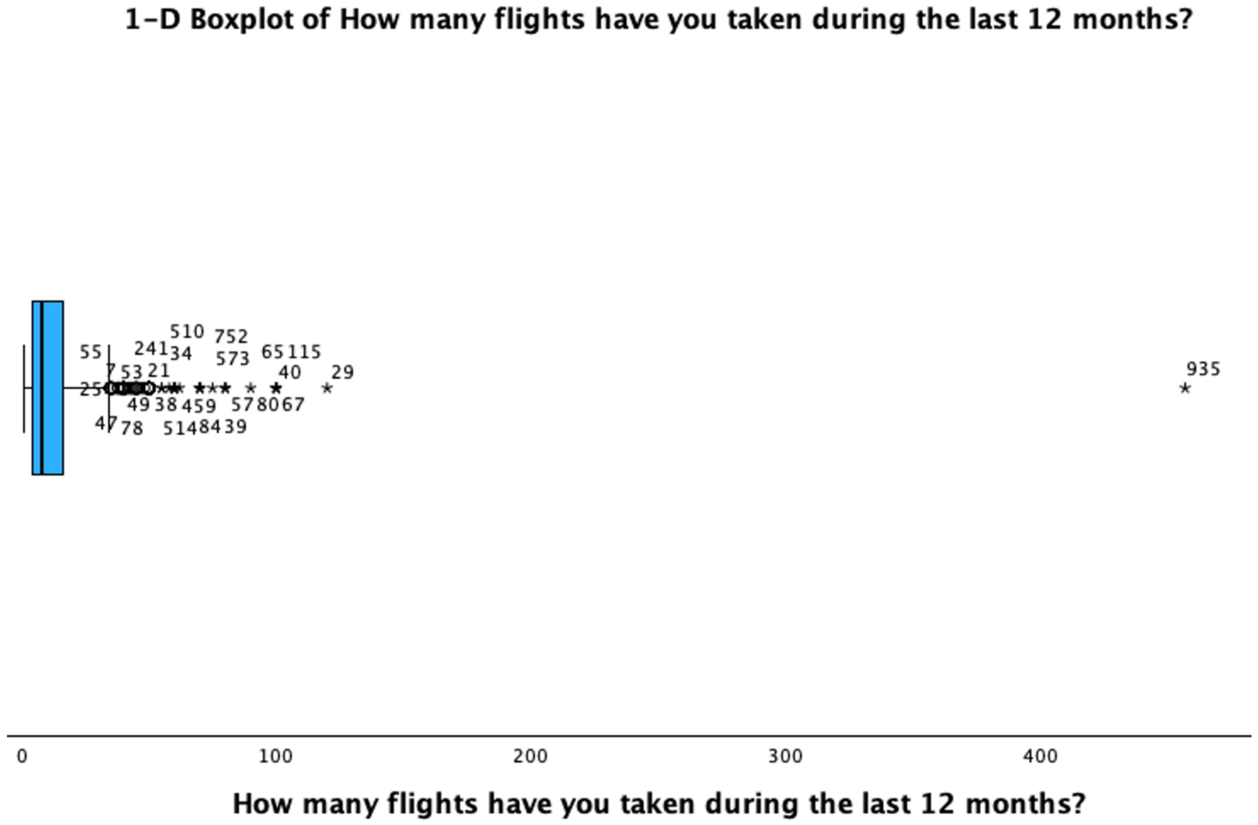
Data Visualization for Continuous Variables

Histogram

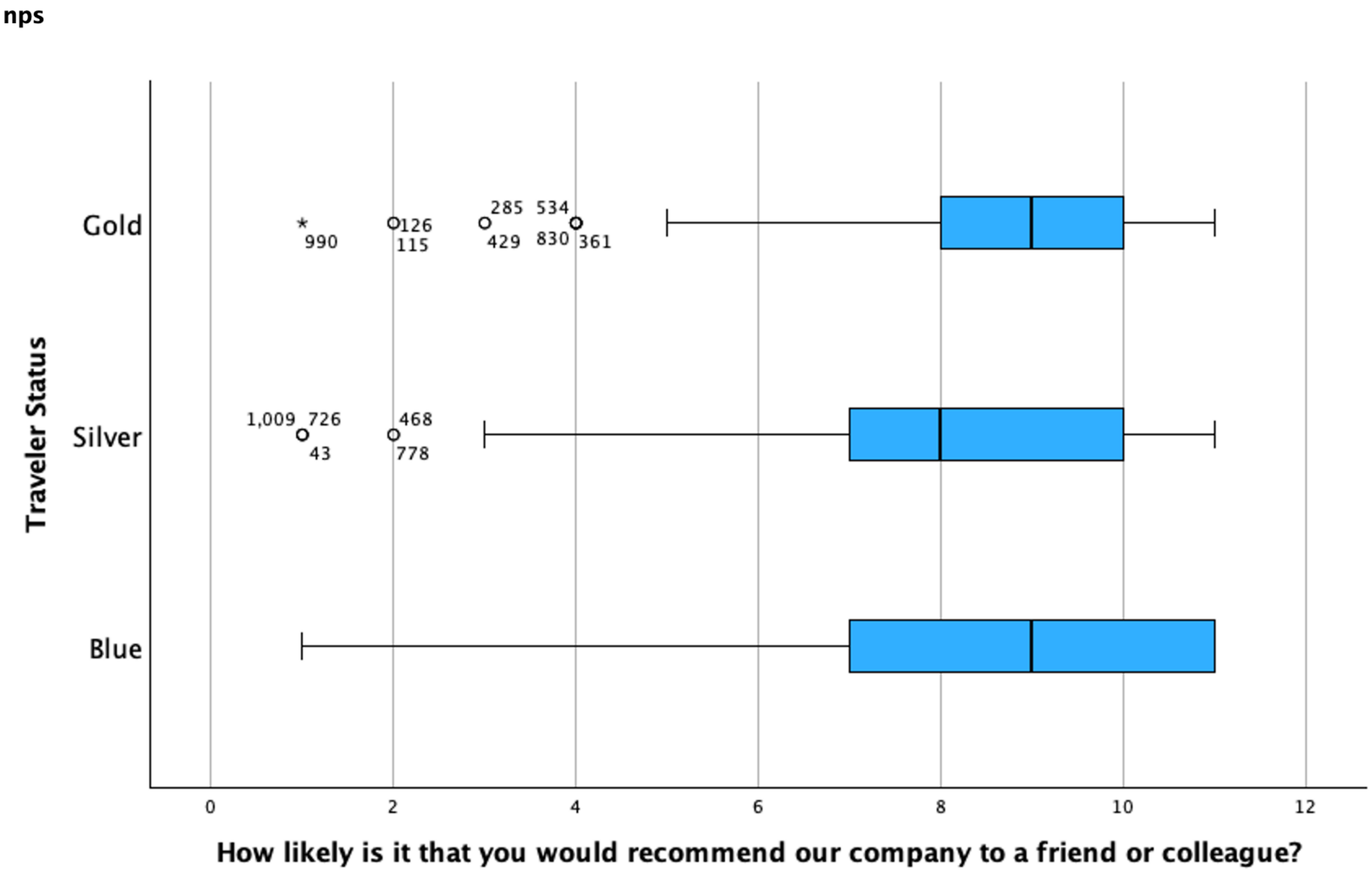


Boxplot

(Basic Element → Transpose for horizontal display)



Data Visualization: Box Plots



Boxplot

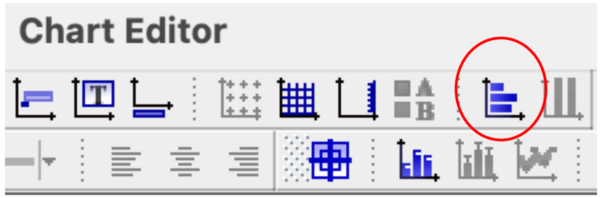
→

Variable – nps

Category axis – status

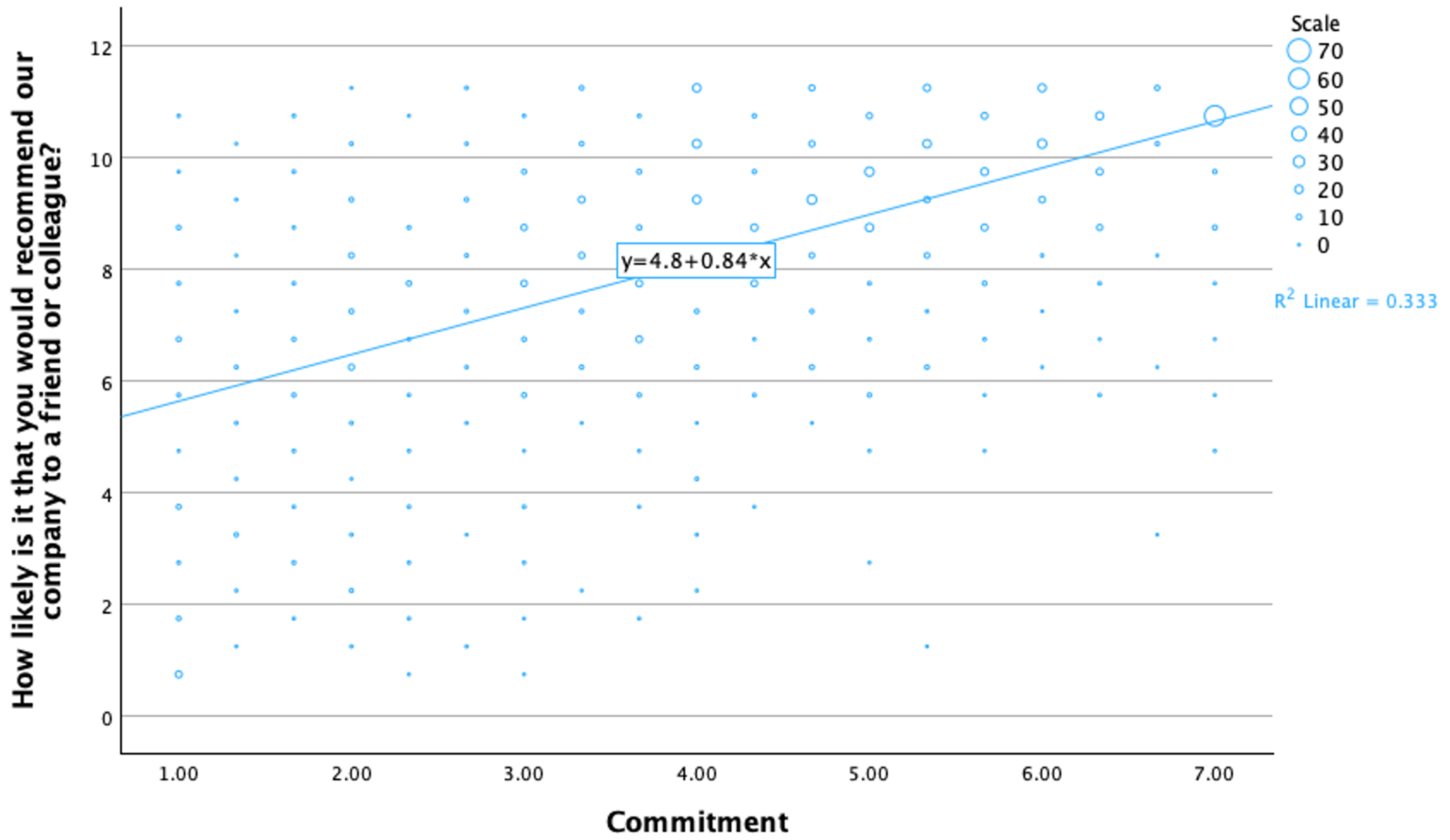
→

Double-click on the chart



Data Visualization: Scatter plot

Graph



Scatter/Dot

→

Simple Scatter

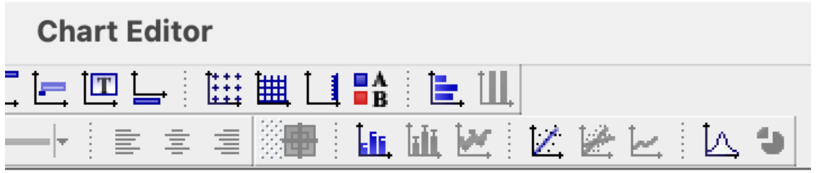
→

Y: nps

X: commitment

→

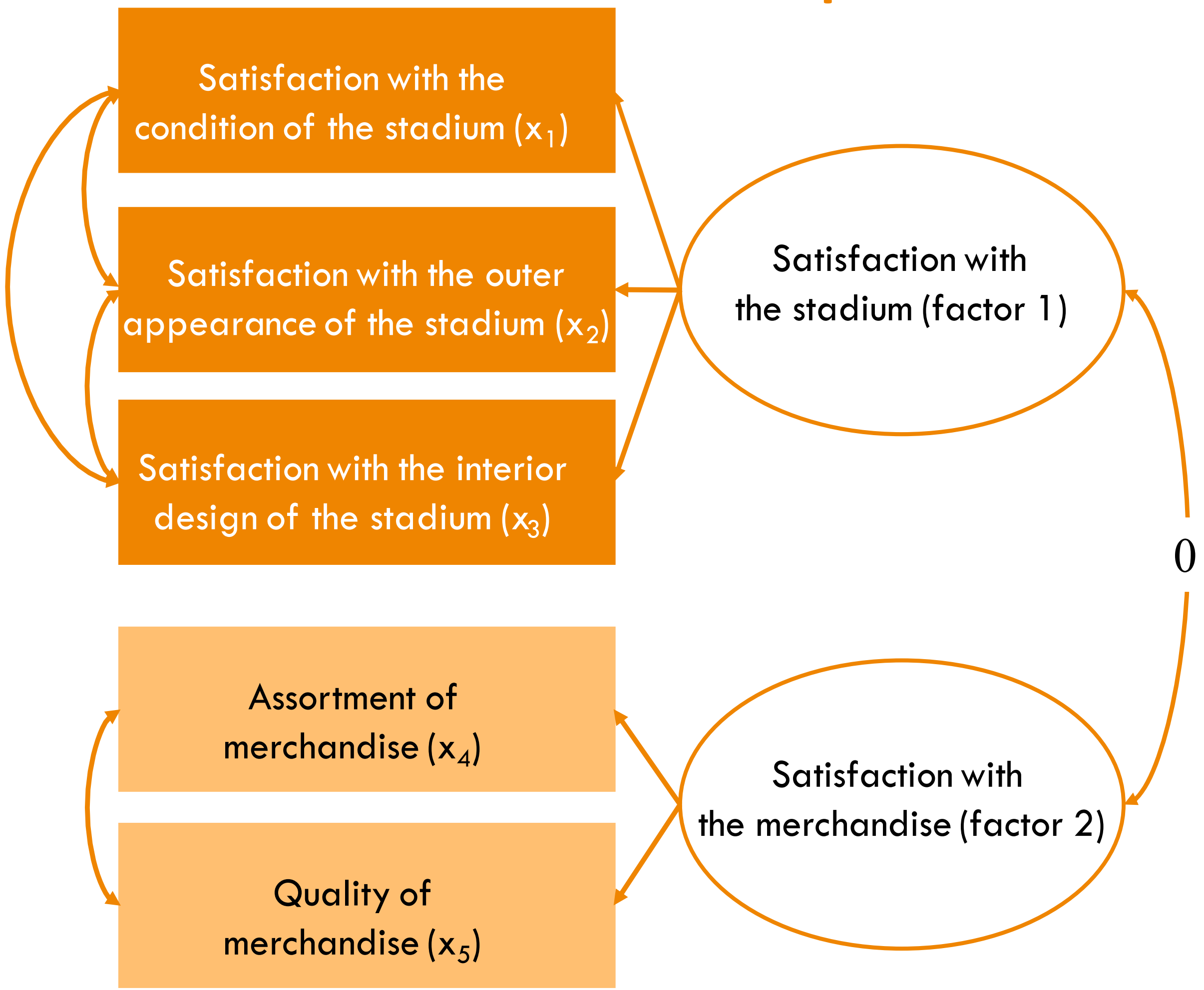
Double-click on the chart



Principal Component & Factor analysis

Concept and Background

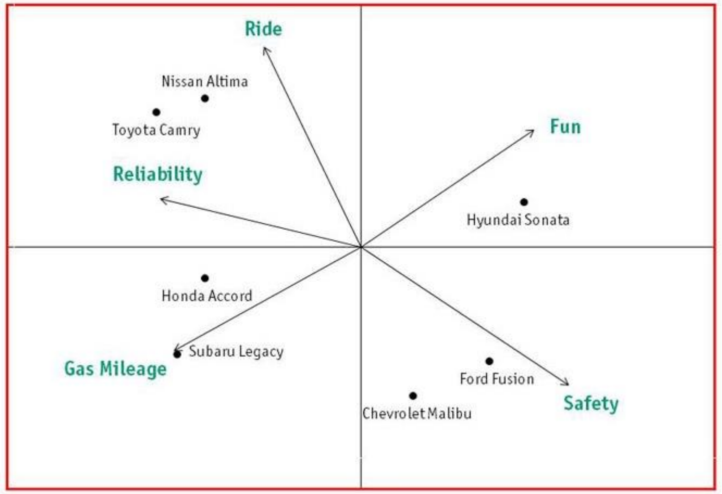
Example: Soccer fan satisfaction



- Assumption: the correlation between the variables is caused by an underlying factor
- The factor is a latent variable which captures the joint meaning of the items related to it
- Example: the variables x_1, x_2, and x_3 can be represented by the factor “Satisfaction with the stadium”

Principal Component Analysis in Marketing

- **Data Reduction**: Condense a large number of correlated variables (e.g., survey items) into fewer uncorrelated components. Example: Reduce 12 attitude items into 2–3 underlying scores (e.g., “brand image”, “perceived value”).
- **Simplification for Interpretation**: Help visualize and interpret high-dimensional data by summarizing it with fewer factors. Example: Mapping car brands by consumer perceptions (e.g., safety, luxury, price).
- **Handling Multicollinearity**: When predictors are correlated (common in survey data), PCA transforms them into orthogonal (uncorrelated) components. This improves model stability.
- **Input for Further Analysis**: PCA component scores can be used in regression analysis (to avoid multicollinearity)
- **Creating Index**: PCA helps create composite indices (e.g., loyalty index, innovativeness score) by weighting survey items used in dashboards and brand tracking studies.



Steps in a Principal Components Analysis

Check requirements and conduct preliminary analyses

Extract the factors

Determine the number of factors

Interpret the factor solution

Evaluate the goodness of fit of the factor solution

Example: Application of PCA

- In the dataset, S1 to S8 asked about different dimensions of satisfaction

Statistics

		N Valid	N Missing	Mean	Median	Std. Deviation
S1	… with Oddjob Airways you will arrive on time.	1038	27	60.91	58.00	26.022
S2	… the entire journey with Oddjob Airways will occur as booked.	1040	25	59.64	56.00	25.750
S3	… in case something does not work out as planned, Oddjob Airways will find a good solution.	954	111	55.62	52.00	25.072
S4	… the flight schedules of Oddjob Airways are reliable.	1035	30	57.27	54.00	27.543
S5	… Oddjob Airways provides you with a very pleasant travel experience.	1041	24	56.61	53.00	22.518
S6	… Oddjob Airways's on board facilities are of high quality.	1041	24	56.21	52.00	22.150
S7	… Oddjob Airways's cabin seats are comfortable.	1048	17	51.76	50.00	24.646
S8	… Oddjob Airways offers a comfortable on-board experience.	1034	31	57.42	53.00	21.402

Positive correlations among s1-s8

Correlations

		s1	s2	s3	s4	s5	s6	s7	s8
s1	Pearson Correlation	--							
	N	1038							
s2	Pearson Correlation	.739**	--						
	Sig. (2-tailed)	<.001							
	N	1037	1040						
s3	Pearson Correlation	.619**	.694**	--					
	Sig. (2-tailed)	<.001	<.001						
	N	952	952	954					
s4	Pearson Correlation	.717**	.766**	.645**	--				
	Sig. (2-tailed)	<.001	<.001	<.001					
	N	1033	1034	951	1035				
s5	Pearson Correlation	.511**	.539**	.559**	.490**	--			
	Sig. (2-tailed)	<.001	<.001	<.001	<.001				
	N	1026	1027	945	1022	1041			
s6	Pearson Correlation	.490**	.498**	.497**	.432**	.821**	--		
	Sig. (2-tailed)	<.001	<.001	<.001	<.001	<.001			
	N	1025	1027	943	1022	1032	1041		
s7	Pearson Correlation	.453**	.455**	.460**	.373**	.787**	.833**	--	
	Sig. (2-tailed)	<.001	<.001	<.001	<.001	<.001	<.001		
	N	1030	1032	947	1027	1038	1037	1048	
s8	Pearson Correlation	.533**	.533**	.554**	.482**	.807**	.840**	.777**	--
	Sig. (2-tailed)	<.001	<.001	<.001	<.001	<.001	<.001	<.001	
	N	1018	1019	937	1014	1029	1025	1032	1034

**. Correlation is significant at the 0.01 level (2-tailed).

Empirical question:
Can we simplify these 8 survey items (for analysis)?

Step 1. Check Requirements and Carry out Preliminary Analyses

Sample Size and Correlations

- Items/ (observable) variables
- Sample size: Is n > 10 • k (n = sample size, k = number of variables)?
- Correlation: Are the variables sufficiently correlated?

How can you decide what “sufficient” actually means?

- Most (bivariate) correlations should be at least 0.30 and significant
- Anti-image as small a possible:

– The **Kaiser-Meyer-Olkin (KMO)** statistic indicates whether the correlations between variables can be explained through the other variables in the dataset (rule of thumb: KMO > 0.50)

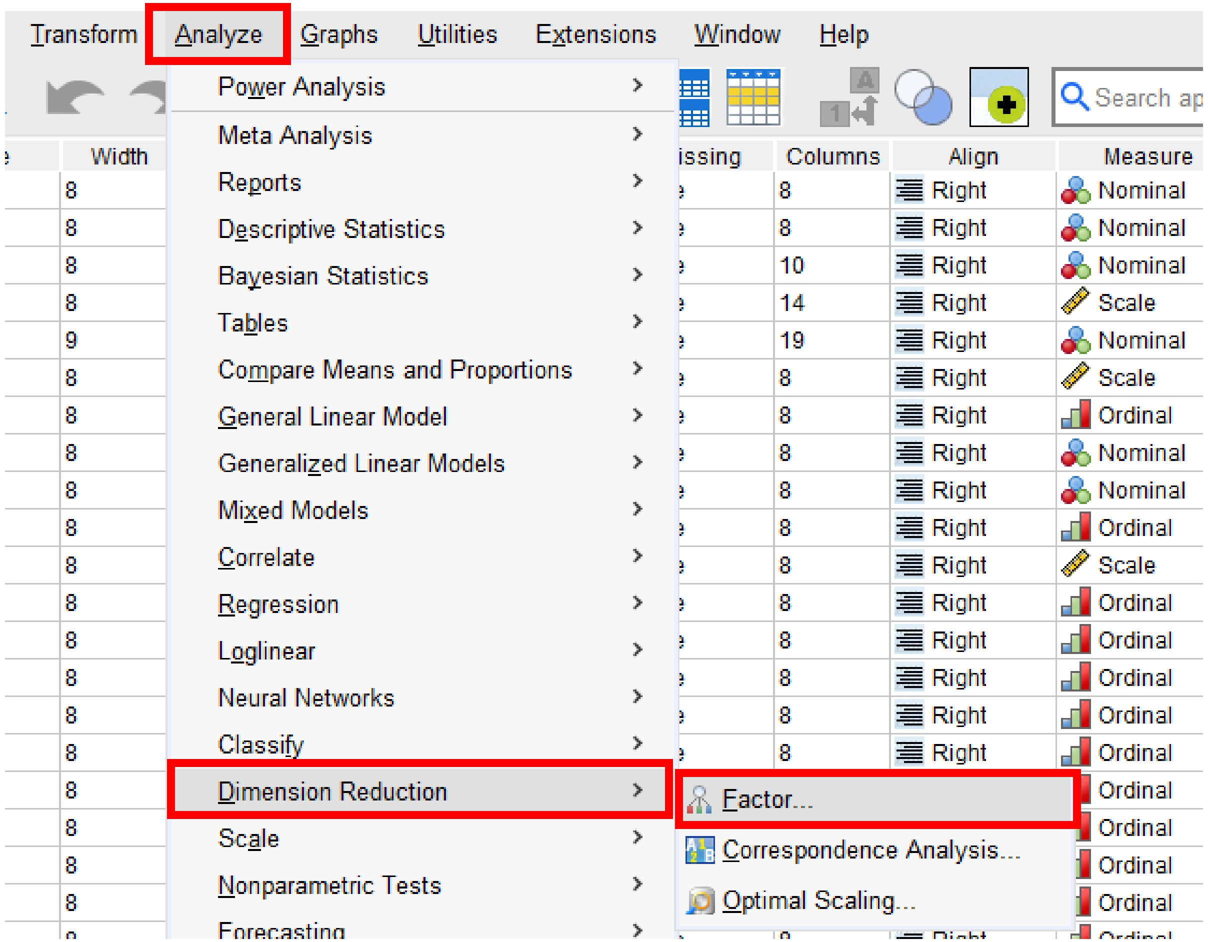
Step 1: Check Assumptions and Carry Out Preliminary Analyses

Example

Use the PCA to reduce the complexity of the item set s1– s8 by extracting a set of uncorrelated factors

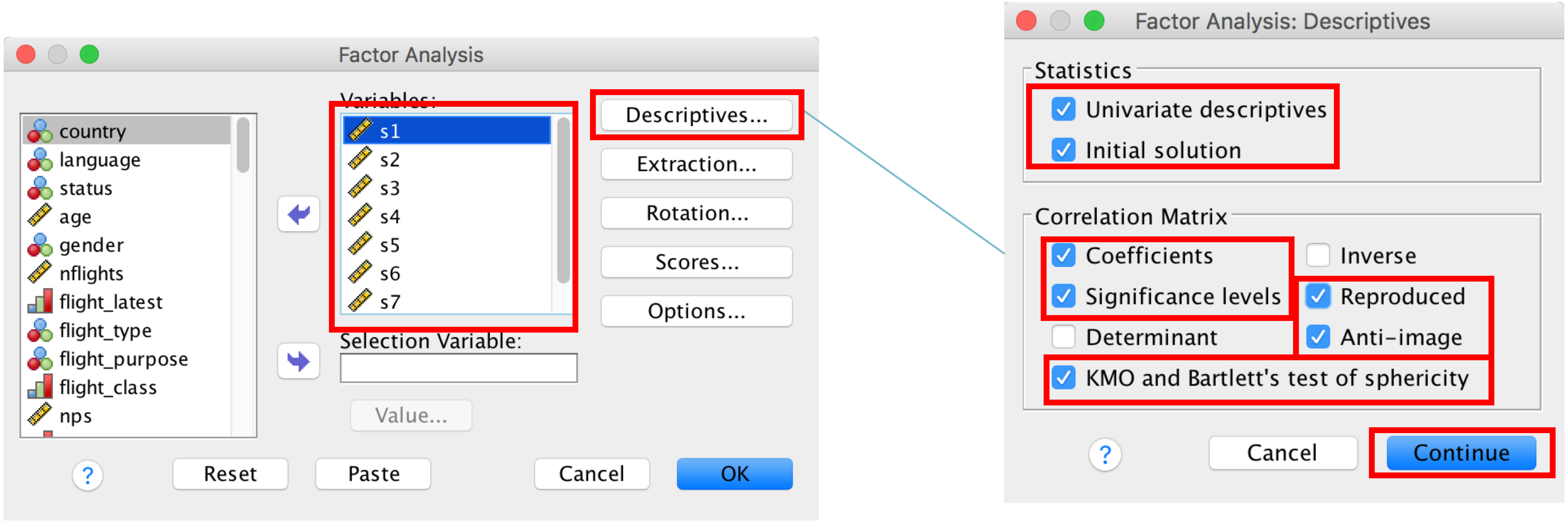
Analyze → Dimension Reduction → Factor → Select variables of interest (s1 ~ s8)

(cont.)



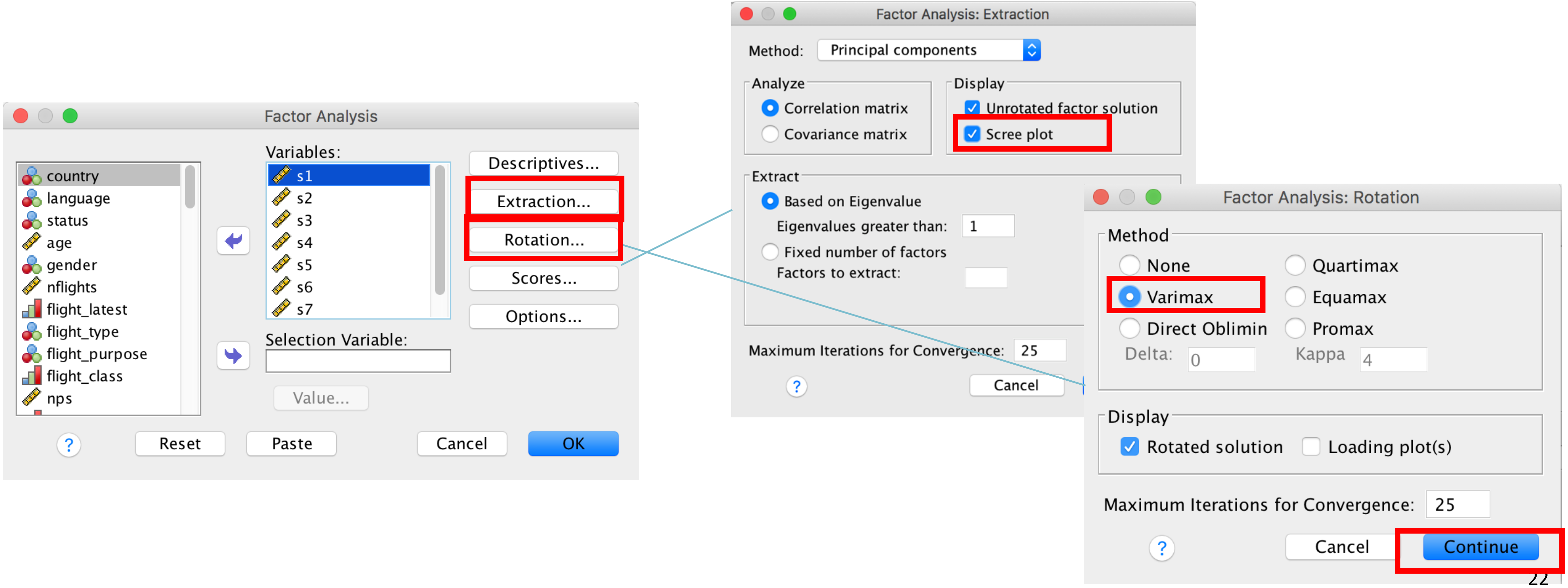
Step 1: Check Assumptions and Carry Out Preliminary Analyses

[Descriptives] → Tick 'Univariate descriptives', 'Coefficients', 'Significance levels', 'Reproduced', 'Anti-image', and 'KMO and Bartlett's test of sphericity' (cont.)



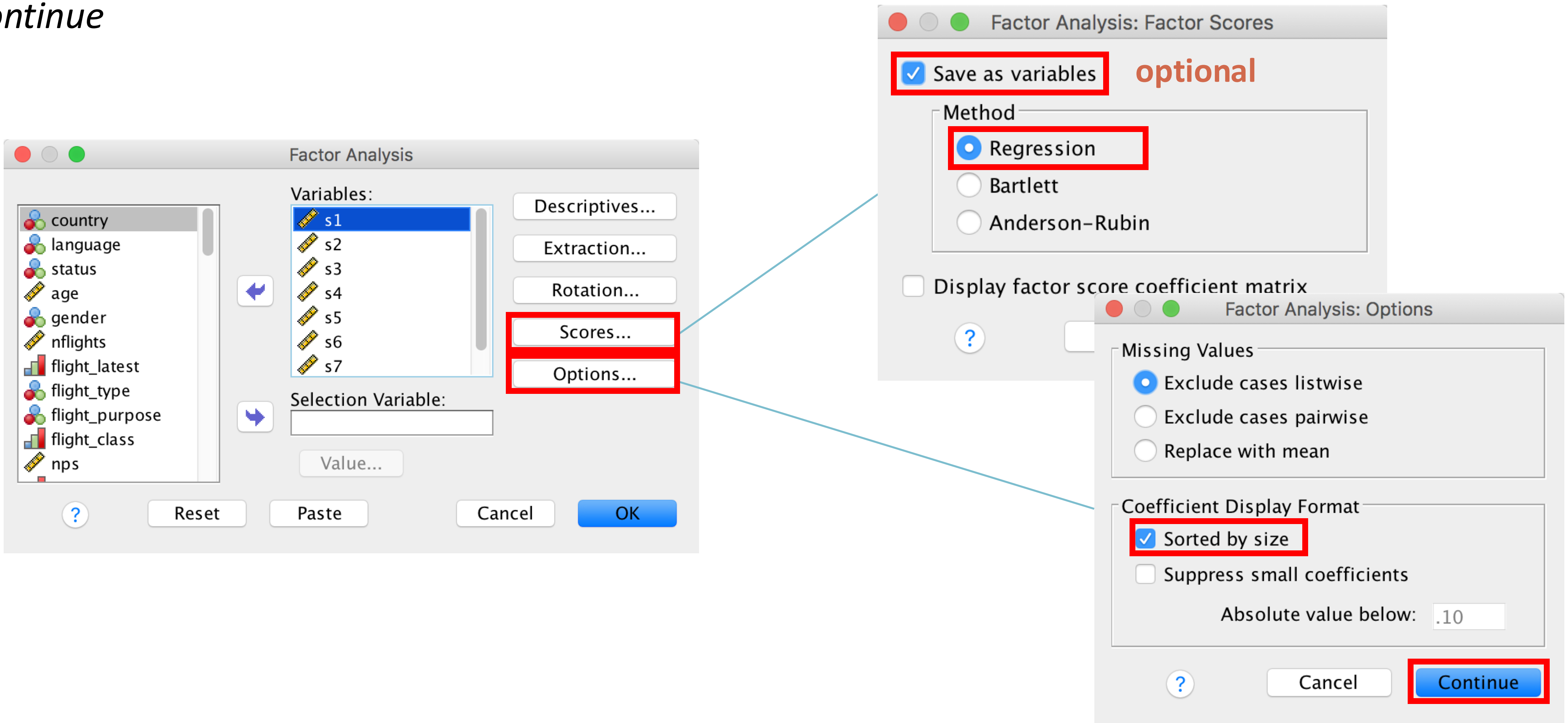
Step 1: Check Assumptions and Carry Out Preliminary Analyses

[Extraction] → Tick 'scree plot' → Continue → [Rotation] → Tick 'Varimax' → Continue → (cont.)



Step 1: Check Assumptions and Carry Out Preliminary Analyses

[Score] → Tick 'Save as variables' and 'Regression' → Continue → [Options] → Tick 'Sorted by size' → Continue



Step 1: Check Assumptions and Carry Out Preliminary Analyses

Descriptive Statistics

	Mean	Std. Deviation	Analysis N
s1	60.36	26.300	921
s2	59.34	25.963	921
s3	55.83	25.150	921
s4	56.89	27.808	921
s5	56.48	22.269	921
s6	56.10	22.070	921
s7	51.60	24.568	921
s8	57.08	21.286	921

Sample size is sufficient

Numerous variables are highly correlated

Correlation Matrix

		s1	s2	s3	s4	s5	s6	s7	s8
		1.000	.754	.622	.733	.525	.500	.455	.542
		.754	1.000	.697	.771	.536	.493	.453	.535
		.622	.697	1.000	.648	.550	.492	.452	.553
		.733	.771	.648	1.000	.484	.436	.367	.487
		.525	.536	.550	.484	1.000	.818	.788	.811
		.500	.493	.492	.436	.818	1.000	.832	.842
		.455	.453	.452	.367	.788	.832	1.000	.780
	s8	.542	.535	.553	.487	.811	.842	.780	1.000
. (1-tailed)	s1		.000	.000	.000	.000	.000	.000	.000
	s2	.000		.000	.000	.000	.000	.000	.000
	s3	.000	.000		.000	.000	.000	.000	.000
	s4	.000	.000	.000		.000	.000	.000	.000
	s5	.000	.000	.000	.000		.000	.000	.000
	s6	.000	.000	.000	.000	.000		.000	.000
	s7	.000	.000	.000	.000	.000	.000		.000
	s8	.000	.000	.000	.000	.000	.000	.000	

Step 1: Check Assumptions and Carry Out Preliminary Analyses

KMO and Bartlett's Test

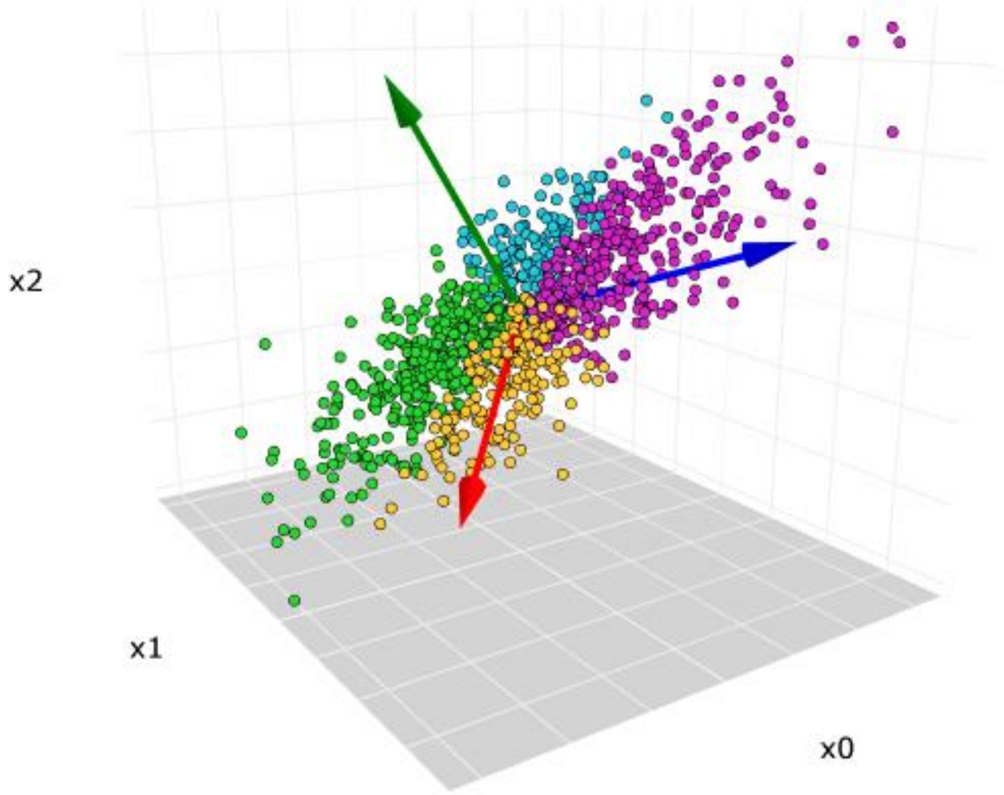
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.907
Bartlett's Test of Sphericity	Approx. Chi-Square	6421.354
	df	28
	Sig.	.000

KMO is “excellent”

rule of thumb: KMO > 0.50

2. Extract the Factors

- Principal component analysis (PCA) is a technique that transforms high-dimensional data into lower dimensions, while retaining as much information as possible.
- PCA: a linear transformation that transforms the data to a new coordinate system such that the greatest variance of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.
- For instance, the red, blue, and green arrows are the direction of the first, second, and third principal components.



2. Extract the Factors

The principal components analysis works with standardized variables

Eigenvalue

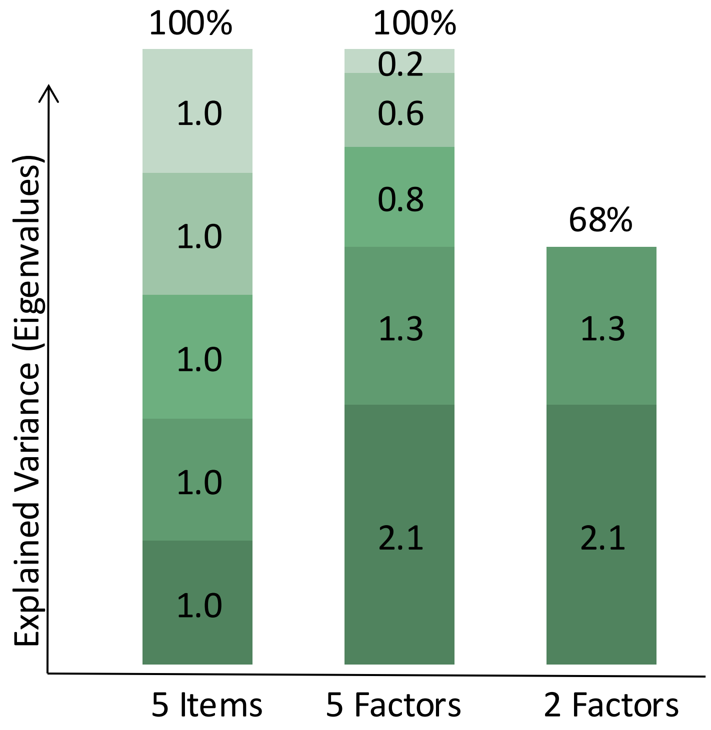
- Describes how much variance is accounted for by a certain factor
- Every additional factor extracted increases the variance accounted for (see picture)

Communality

- Describes how much of each variable's variance is captured or reproduced by the factors extracted

Factor loadings

- Correlations between the factors and the variables



Steps 2: Extract the Factors

The two factors with eigenvalues jointly explain 82.215% of the total variance

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.249	65.611	65.611	5.249	65.611	65.611	3.411	42.633	42.633
2	1.328	16.604	82.215	1.328	16.604	82.215	3.167	39.582	82.215
3	.397	4.966	87.181						
4	.263	3.290	90.471						
5	.231	2.890	93.361						
6	.196	2.454	95.815						
7	.193	2.409	98.224						
8	.142	1.776	100.000						

Extraction Method: Principal Component Analysis.

Extracting as many factors as variables leads to full information retrieval

Step 3. Determine the Number of Factors

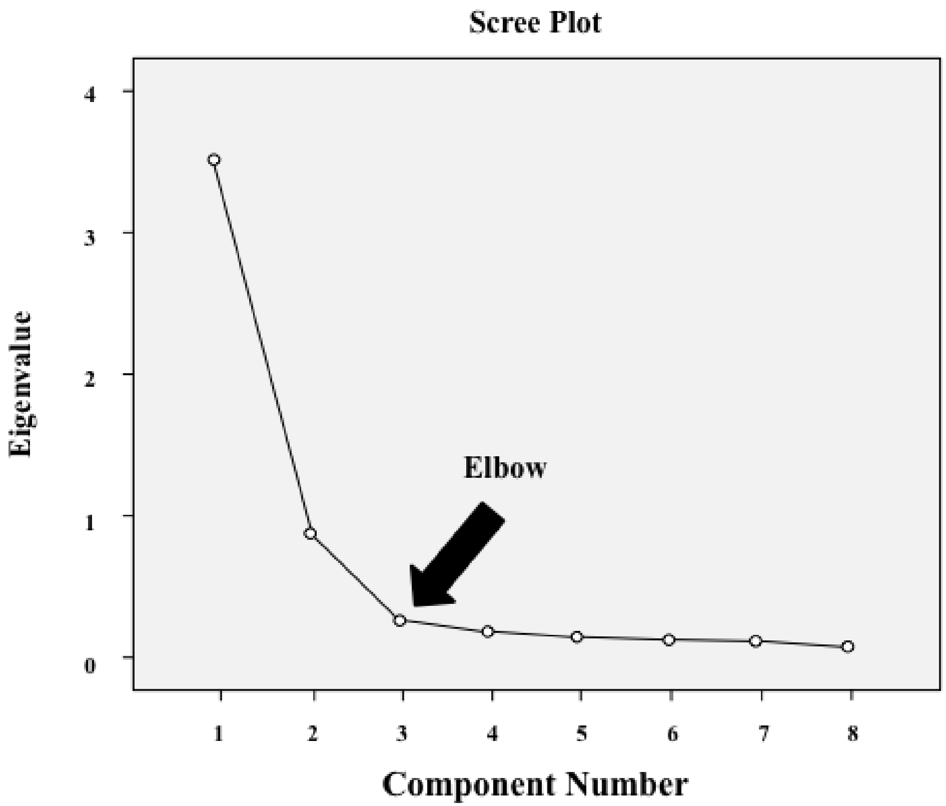
The number of factors can be determined in different ways:

Option 1: Kaiser criterion

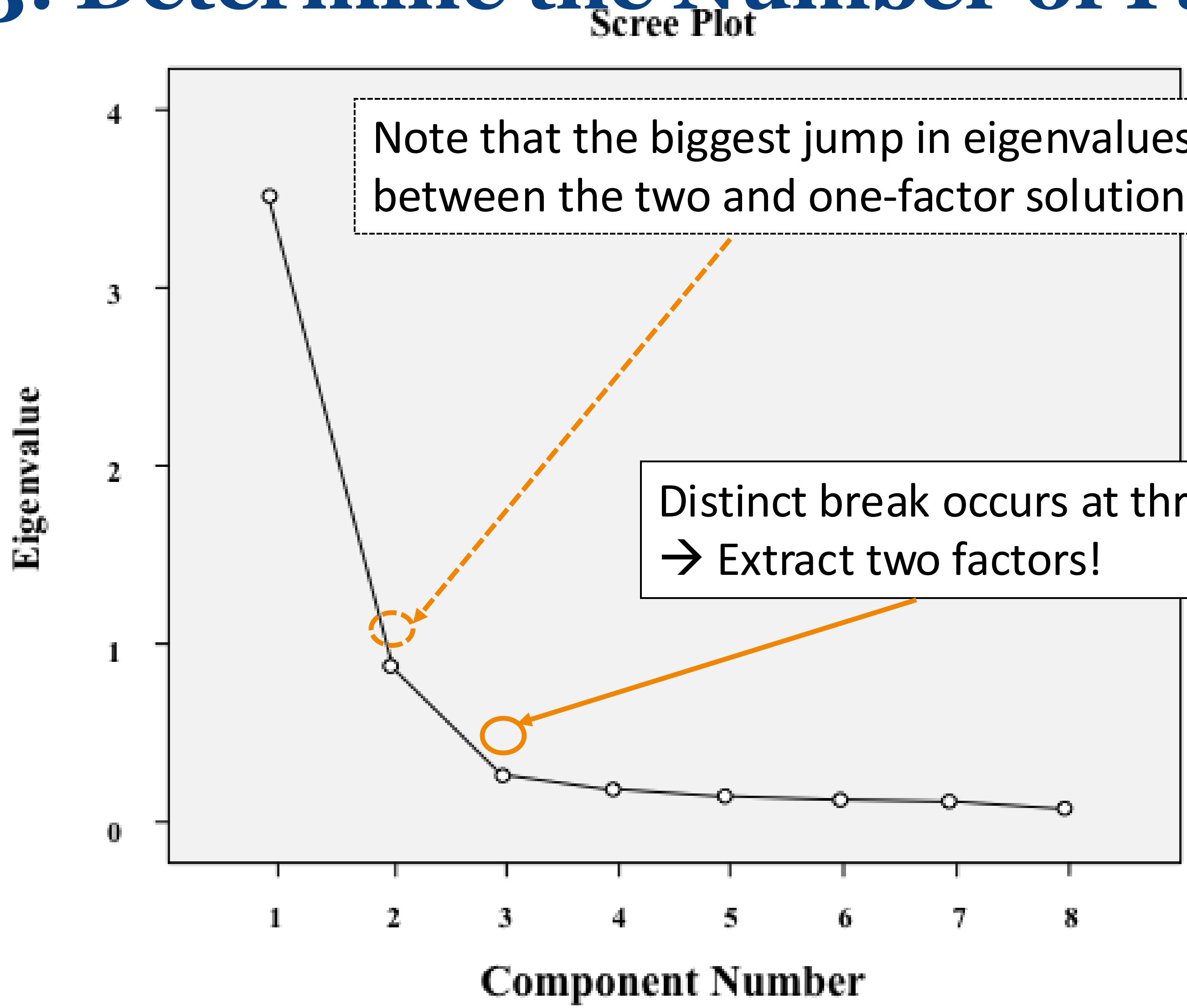
- Extract all factors with an Eigenvalue > 1
- Every factor extracted describes more than one variable

Option 2: Scree Plot**

- Distinct break (elbow) occurs for 2 factors
  → Extract one factor less than indicated by elbow



Step 3: Determine the Number of Factors



Note that the biggest jump in eigenvalues always occurs between the two and one-factor solution

Distinct break occurs at three-factor solution
→ Extract two factors!

Steps 2: Extract the Factors

The two factors with eigenvalues jointly explain 82.215% of the total variance

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.249	65.611	65.611	5.249	65.611	65.611	3.411	42.633	42.633
2	1.328	16.604	82.215	1.328	16.604	82.215	3.167	39.582	82.215
3	.397	4.966	87.181						
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8	.142	1.776	100.000						

Extraction Method: Principal Component Analysis.

Extracting as many factors as variables leads to full information retrieval

Only the first two factors have eigenvalues greater than one

4. Interpret the Factor Solution

- The factor loadings vary in a range from -1 (strong negative influence) to +1 (strong positive influence)
- Low absolute value of factor loadings (near 0): The variable is represented badly by the factor

Rules of thumb

- Typically every variable is "assigned" to the factor with the highest loadings
- To label the factor, find an "umbrella term" for the variables "assigned" to it

Rotated Component Matrix[a]

	Component	
	1	2
s6	.903	.272
s7	.902	.208
s8	.856	.352
s5	.852	.347
s4	.198	.885
s2	.282	.871
s1	.299	.829
s3	.340	.759

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 3 iterations.

Steps 4: Interpret the Factor Solution

Variables s5 – s8 load highly on the first factor

→ Satisfaction with the onboard experience

Rotated Component Matrix[a]

	Component	
	1	2
... Oddjob Airways's on board facilities are of high quality. S6	.903	.272
... Oddjob Airways's cabin seats are comfortable. S7	.902	.208
... Oddjob Airways offers a comfortable on-board experience. S8	.856	.352
... Oddjob Airways provides you with a very pleasant travel experience. S5	.852	.347
... the flight schedules of Oddjob Airways are reliable. S4	.198	.885
... the entire journey with Oddjob Airways will occur as booked. S2	.282	.871
... with Oddjob Airways you will arrive on time. S1	.299	.829
... in case something does not work out as planned, Oddjob Airways will find a good solution. S3	.340	.759

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 3 iterations.

Variables s1 – s4 load highly on the second factor

→ Satisfaction with the reliability

Step 5. Evaluate the Goodness-of-fit of the Factor Solution

- Overall objective: extract factors in such a way that the factors and their loadings reproduce the data in the best possible way
- Look into the differences between the initial and reproduced correlation matrices.

1. Check the proportion of the correlation matrices' residuals with an absolute value higher than 0.05.

2. **Check the variables' communalities****
   Communality is less than 0.3
   → Reconsider your set-up

Communalities

	Initial	Extraction
s1	1.000	.777
s2	1.000	.838
s3	1.000	.691
s4	1.000	.822
s5	1.000	.847
s6	1.000	.890
s7	1.000	.856
s8	1.000	.856

Extraction Method: Principal Component Analysis.

Steps 5: Evaluate the Goodness-of-fit

- Overall objective: extract factors in such a way that the factors and their loadings reproduce the data in the best possible way
- Look into the differences between the initial and reproduced correlation matrices.
- **Check the variables' communalities.** If communality is less than 0.3
  → Reconsider your set-up
- Check the proportion of the correlation matrices' residuals with an absolute value higher than 0.05.

All communalities are high, providing support for the solution's goodness-of-fit

Communalities

	Initial	Extraction
... with Oddjob Airways you will arrive on time. S1	1.000	.777
... the entire journey with Oddjob Airways will occur as booked. S2	1.000	.838
... in case something does not work out as planned, Oddjob Airways will find a good solution. S3	1.000	.691
... the flight schedules of Oddjob Airways are reliable. S4	1.000	.822
... Oddjob Airways provides you with a very pleasant travel experience. S5	1.000	.847
... Oddjob Airways's on board facilities are of high quality. S6	1.000	.890
... Oddjob Airways's cabin seats are comfortable. S7	1.000	.856
... Oddjob Airways offers a comfortable on-board experience. S8	1.000	.856

Extraction Method: Principal Component Analysis.

Workshop Activity 1

Use the PCA to reduce the complexity of the item set – s6, s7, s8, s9, … s12, s13 – by extracting a set of uncorrelated factors

Analyze → Dimension Reduction → Factor → Select variables of interest [Descriptives] → Tick 'Univariate descriptives', 'Coefficients', 'Significance levels', 'Reproduced', 'Anti-image', and 'KMO and Bartlett's test of sphericity' → [Extraction] → Tick 'scree plot' → Continue → [Rotation] → Tick 'Varimax' → Continue → [Score] → Tick 'Save as variables' and 'Regression' → Continue → [Options] → Tick 'Sorted by size' → Continue

1. How many factors are identified?
2. Name the factors.

Appendix

Concept and Background

Factor analysis

Definition

Identification of latent variables (i.e., variables that are not directly observable)

▶ *Analysis and interpretation of the correlation between the variables in the data set*

Goals

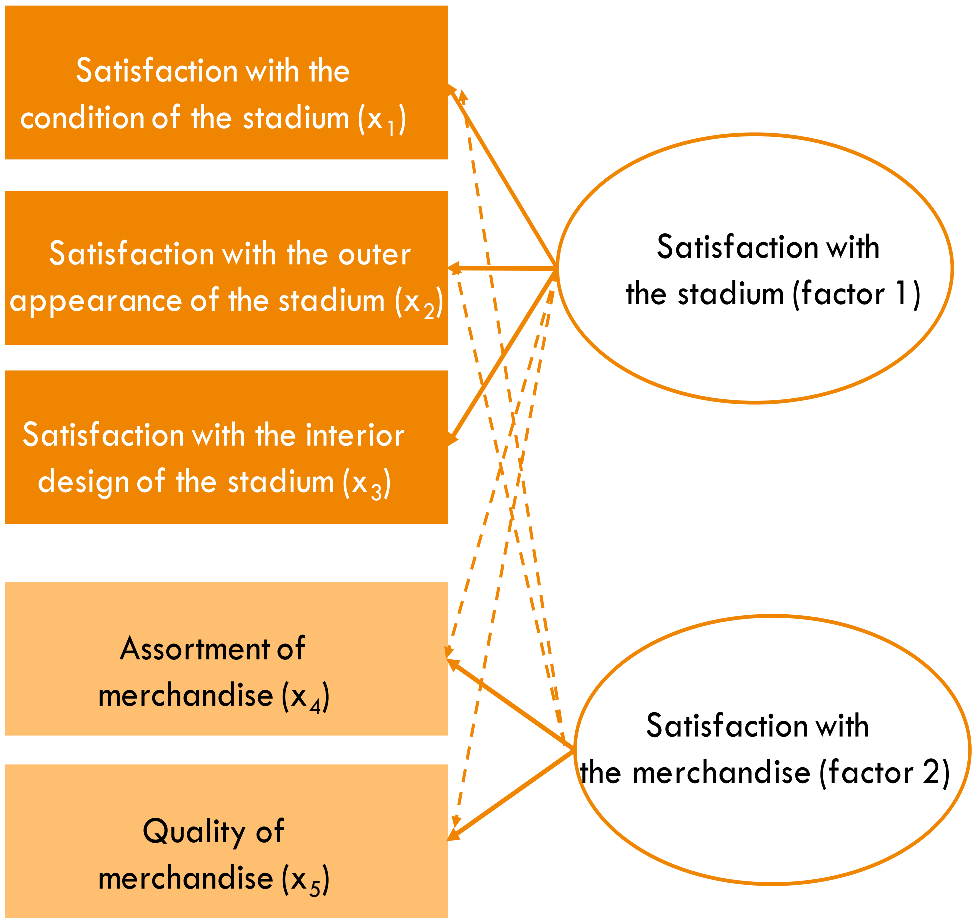
- Data reduction
- Generate hypothesis regarding the composition of factors
- Screen items for subsequent analysis

Types

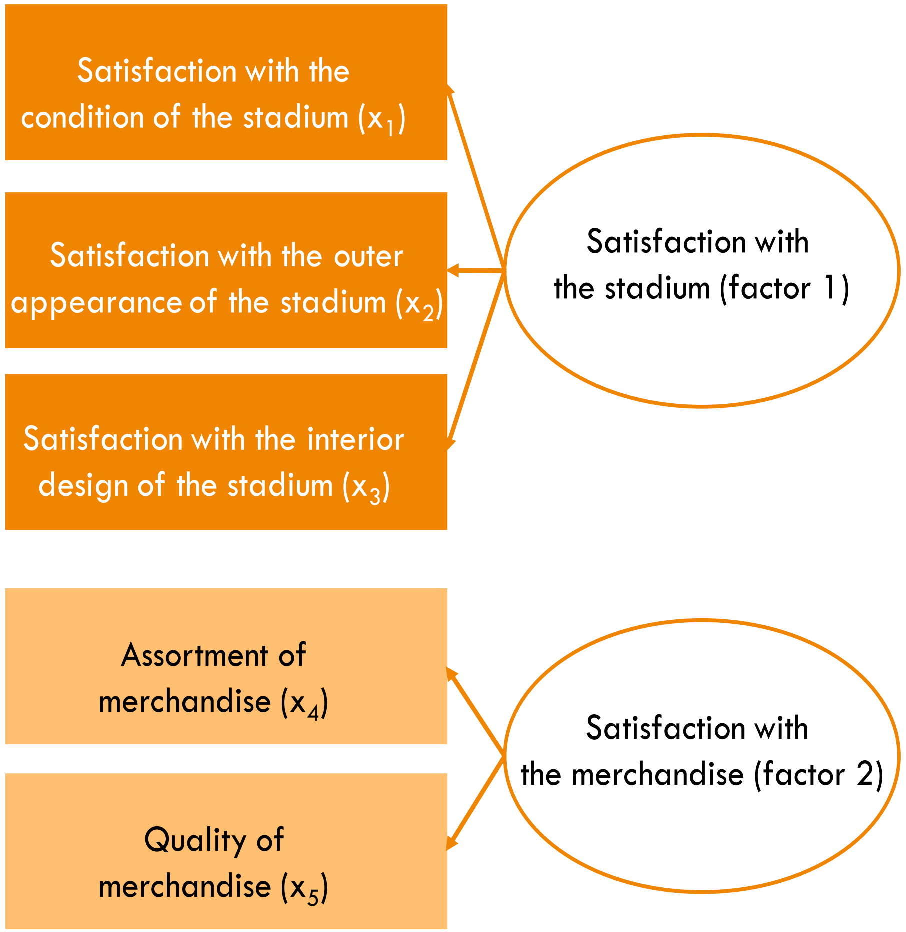
- Exploratory factor analysis: What's the underlying structure?
  - Principal component analysis (PCA)
  - Principal axis factoring
- Confirmatory factor analysis: Do my data fit the hypothesized structure?

Explanatory vs Confirmatory Factor Analysis

Example: Soccer fan satisfaction



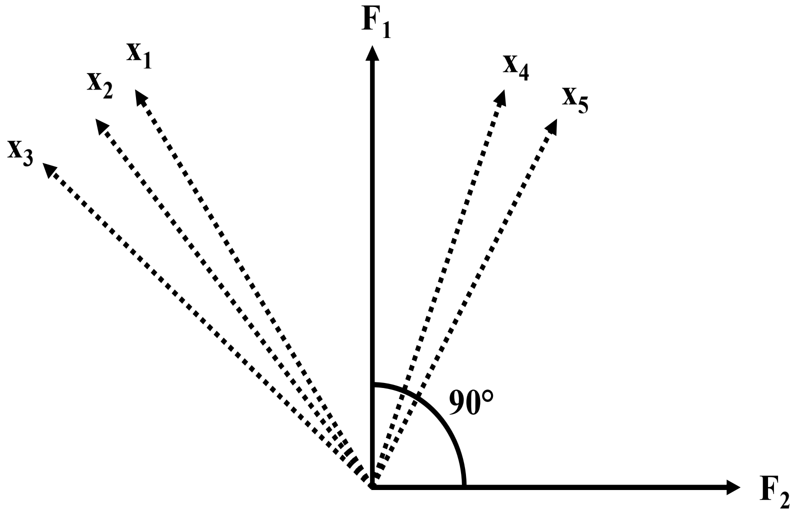
Exploratory factor analysis



Confirmatory factor analysis

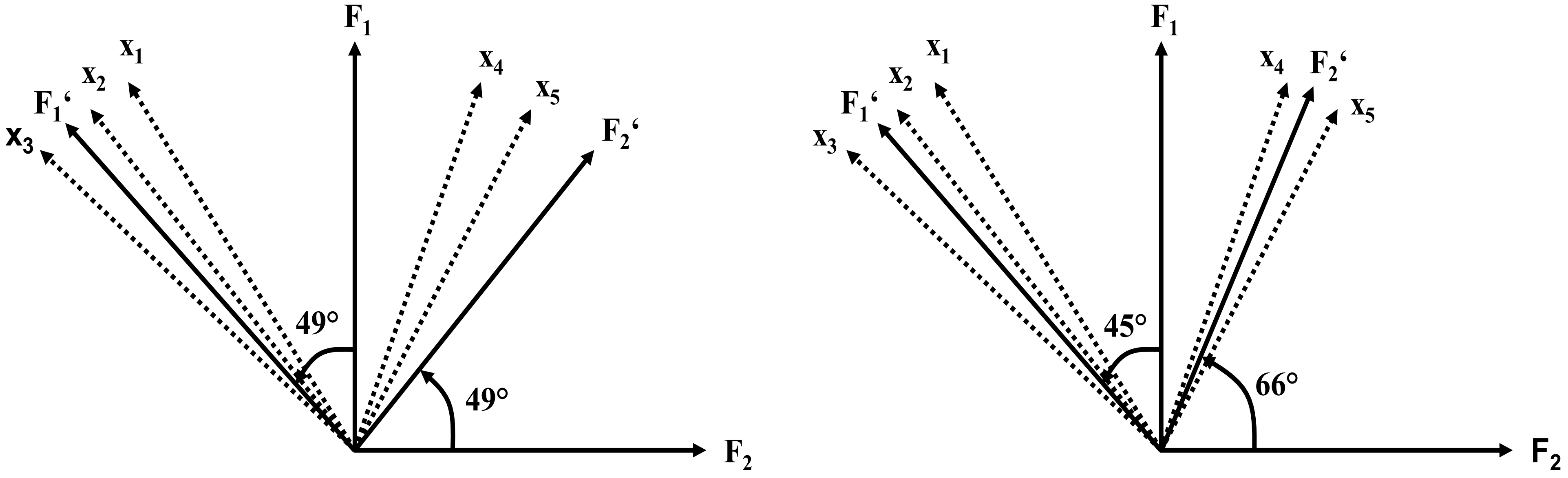
Step 2. Extract the Factors

- Each extracted factor maximizes the variance accounted for
- Factors are extracted in such a way that the initial correlation matrix of the variables is reproduced in the best possible way
- The vectors x1-x5 each represent one variable
- The angles between the vectors can be interpreted as correlations
- The first factor (F1) is fitted into this vector space in such a way that the sum of all the angles with this factor is minimized
- After this, a second factor (F2) is extracted which maximizes the remaining variance accounted for
- This factor is independent from the first one and, thus, is fitted into the vector space at a 90° angle



Step 4. Interpret the Factor Solution

The **Varimax** procedure is the most prominent orthogonal rotation method



Orthogonal factor rotation
(e.g., varimax*)

Oblique factor rotation
(e.g., promax rotation**)

*default option for orthogonal rotation in SPSS

**recommended oblique rotation method (set Kappa=3)